IP Addressing & Subnetting

Task 1 : List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

An IPv4 address is a 32-bit address divided into five traditional classes: A, B, C, D, and E. Each class has a different purpose.

IP Class	First Octet Range	Default Subnet Mask	Purpose	Real-World Example
Class A	1–127	255.0.0.0 (/8)	Historically used for very large networks	A large organization's private network using 10.0.0.0/8
Class B	128–191	255.255.0.0 (/16)	Historically used for medium-to-large networks	A college network using private addresses from 172.16.0.0/16
Class C	192–223	255.255.255.0 (/24)	Historically used for smaller networks	A home LAN using 192.168.1.0/24
Class D	224–239	Not applicable	Multicast communication	A router or streaming service using an IPv4 multicast group address
Class E	240–255*	Not applicable	Reserved for experimental or future use	Research and testing involving reserved IPv4 address ranges

Important Notes

1. Class A addresses traditionally start at 1; 127.0.0.0/8 is reserved for loopback.
2. Private IPv4 ranges include 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.
3. Class D is used for multicast, not ordinary host addressing.
4. Class E is reserved and is not normally used for regular devices.
5. Modern networks primarily use CIDR (Classless Inter-Domain Routing), which allows flexible subnet masks instead of relying on traditional address classes.

Task 2 : Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Hint: Show your working steps for each calculation.

Given Information

- IP Address: 192.168.10.45
- Subnet Mask: 255.255.255.0
- CIDR Notation: /24

Step 1: Identify the Network Portion

The subnet mask 255.255.255.0 means the first three octets represent the network portion, and the last octet represents the host portion.

IP Address: 192.168.10.45
Subnet Mask: 255.255.255.0

Step 2: Calculate the Network Address

Set all host bits to 0.

Network Address: **192.168.10.0**

Step 3: Calculate the Broadcast Address

Set all host bits to 1.

Broadcast Address: **192.168.10.255**

Step 4: Calculate the Valid Host Range

The first usable host address is the network address plus 1.

First Valid Host: **192.168.10.1**

The last usable host address is the broadcast address minus 1.

Last Valid Host: **192.168.10.254**

Step 5: Calculate the Number of Valid Hosts

A /24 network has 8 host bits.

Total addresses = 2^8 = 256

Usable host addresses = 256 − 2 = **254**

Two addresses are excluded: the network address and the broadcast address.

Final Answer

Parameter	**Result**
IP Address	192.168.10.45
Subnet Mask	255.255.255.0
CIDR Prefix	/24
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
First Valid Host	192.168.10.1
Last Valid Host	192.168.10.254
Total Usable Hosts	254

Task 3 : Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Task 3: Subnetting Plan for a Food Delivery Startup

Objective

A food delivery startup like Zomato requires four separate subnets for Delivery Partners, Restaurants, Customers, and Admins. The given network is 10.0.0.0/24.

Step 1: Calculate the New Subnet Mask

Given network: 10.0.0.0/24

Required subnets: 4

Number of borrowed bits:

2^2 = 4 subnets

New prefix length:

/24 + 2 = /26

New subnet mask: **255.255.255.192**

Addresses per subnet:

2^6 = 64 total addresses

Usable hosts per subnet:

64 − 2 = **62 usable hosts**

Step 2: Subnet Allocation Table

No.	Department	Network Address	Subnet Mask	First Valid Host	Last Valid Host	Broadcast Address
1	Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1	10.0.0.62	10.0.0.63
2	Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65	10.0.0.126	10.0.0.127

3	Customers	10.0.0.128/26	255.255.255.192	10.0.0.129	10.0.0.190	10.0.0.191
4	Admins	10.0.0.192/26	255.255.255.192	10.0.0.193	10.0.0.254	10.0.0.255

Step 3: Explanation

The original /24 network is divided into four equal /26 subnets by borrowing two host bits. Each subnet contains 64 total IP addresses, of which 62 are usable for devices. Each department receives a separate subnet, helping organize network resources and apply access controls.

Task 4 : You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Given Information

- Network Address: 172.16.0.0/16
- Required Subnets: At least 20
- Objective: Calculate the new subnet mask and usable hosts per subnet.

Step 1: Calculate the Number of Borrowed Bits

The formula for calculating the number of subnets is:

Number of Subnets = 2^n

Where n is the number of borrowed host bits.

We need at least 20 subnets:

- 2^4 = 16 subnets (not enough)
- **2^5 = 32 subnets (sufficient)**

Therefore, we must borrow **5 bits**.

Step 2: Calculate the New Prefix Length

Original prefix length = /16

Borrowed bits = 5

New prefix length = 16 + 5 = **/21**

Step 3: Calculate the New Subnet Mask

The new prefix is /21.

Subnet Mask: **255.255.248.0**

Binary representation:

11111111.11111111.11111000.00000000

Step 4: Calculate the Number of Hosts per Subnet

IPv4 addresses contain 32 bits.

Remaining host bits = 32 − 21 = 11

Total addresses per subnet = 2^{11} = 2048

Usable hosts per subnet = 2048 − 2 = **2046**

Two addresses are reserved for the network address and broadcast address.

Step 5: Final Answers

Parameter	**Result**
Original Network	172.16.0.0/16
Required Subnets	At least 20
Borrowed Bits	5
Total Subnets Created	32
New CIDR Prefix	/21
New Subnet Mask	255.255.248.0
Host Bits Remaining	11
Total Addresses per Subnet	2048
Usable Hosts per Subnet	2046

Task 5 : A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

Given Information

IP Address: 192.168.1.300

1. Identify the Error

The IP address 192.168.1.300 is invalid because an IPv4 address consists of four octets, and each octet must have a value between 0 and 255.

In this address, the last octet is 300, which exceeds the maximum allowed value of 255. Therefore, the device cannot use this as a valid IPv4 address.

2. Provide a Valid IP Address

Assuming the network uses the subnet mask 255.255.255.0 (/24), a valid host IP address could be:

192.168.1.100

Other valid host addresses in this subnet range from 192.168.1.1 to 192.168.1.254, provided the address is not already assigned to another device.

- Network Address: 192.168.1.0
- Subnet Mask: 255.255.255.0
- Broadcast Address: 192.168.1.255
- Valid Host Range: 192.168.1.1 – 192.168.1.254

3. Conclusion

The device cannot use 192.168.1.300 because an IPv4 octet cannot exceed 255. Replacing it with a valid, unused address such as 192.168.1.100, along with the correct subnet mask and gateway settings, can resolve the addressing error.